

VECTOR SPACES

A vector space $\mathcal{V}$, defined over a field F , is a non-empty set (whose members are called vectors) equipped with two operations :-

(i) Vector addition, $+$: $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$

(ii) Scalar Multiplication, $\cdot$: $F \times \mathcal{V} \rightarrow \mathcal{V}$

which must satisfy the following properties

(0) Closure under Vector Addition and Scalar Multiplication:

(follows from the way these operations are defined)

If $u, v \in \mathcal{V}$, then $u+v \in \mathcal{V}$, and $\alpha \in F, u \in \mathcal{V}, \alpha \cdot u \in \mathcal{V}$

(i) $u+v = v+u$, $\forall u, v \in \mathcal{V}$ (Commutativity of vector addition)

(ii) $(u+v)+w = u+(v+w)$, $\forall u, v, w \in \mathcal{V}$ (Associativity)

(iii) Existence of zero: There is a vector $0 \in \mathcal{V}$ such that $0+v = v$, $\forall v \in \mathcal{V}$

(iv) Existence of additive inverse: Every $v \in \mathcal{V}$ has another element $v_I \in \mathcal{V}$ such that $v+v_I = 0$

Field Multiplication

(v) $\alpha \cdot (\beta \cdot v) = (\alpha \cdot \beta) \cdot v$, $\forall \alpha, \beta \in F, v \in \mathcal{V}$ (Associativity of scalar multiplication)

(vi) $1 \cdot v = v$, $1 \in F, v \in \mathcal{V}$

(vii) $\alpha \cdot (u+v) = \alpha \cdot u + \alpha \cdot v$ (viii) $(\alpha+\beta) \cdot v = \alpha \cdot v + \beta \cdot v$
 $\forall \alpha \in F, u, v \in \mathcal{V}$ $\alpha, \beta \in F, v \in \mathcal{V}$
 Distributivity.

Examples

① Let F be a field. Define

$$F^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, x_i \in F \right\}$$

ordered list

Vector addition: $x, y \in F^n$

Define:

$$x + y =$$

$$\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

field addition

Scalar multiplication: $\alpha \in F, x \in F^n$

$$\alpha \cdot x = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}$$

Field multiplication

F^n is a vector space with these operations

Zero:

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\leftarrow 0 \in F$. repeated n times

$$x_I = \begin{bmatrix} -x_1 \\ \vdots \\ -x_n \end{bmatrix}$$

$-x_i$ is the additive inverse of $x_i \in F$.

F^n is a Vector Space over F

special cases
of
 F^n

(i) $F = \mathbb{R}$
 $F^n = \mathbb{R}^n$

n -dimensional real Euclidean space.

(ii) $F = \mathbb{C}$
 $F^n = \mathbb{C}^n$

② Let F be a field. Define

$$F^{m \times n} = \left\{ \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & \dots & \dots & x_{mn} \end{bmatrix} \mid x_{ij} \in F \right\}$$

Vector Addition: $X, Y \in F^{m \times n}$

$$X + Y = \begin{bmatrix} x_{11} + y_{11} & \dots & x_{1n} + y_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} + y_{m1} & \dots & x_{mn} + y_{mn} \end{bmatrix}$$

Scalar Multiplication: $\alpha \in F, X \in F^{m \times n}$

$$\alpha X = \begin{bmatrix} \alpha x_{11} & \alpha x_{12} & \dots & \alpha x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha x_{m1} & \dots & \dots & \alpha x_{mn} \end{bmatrix}$$

Zero :

$$\begin{bmatrix} 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & 0 \end{bmatrix}$$

X_I :

$$\begin{bmatrix} -x_{11} & \dots & -x_{1n} \\ \vdots & \ddots & \vdots \\ -x_{m1} & \dots & -x_{mn} \end{bmatrix}$$

$-x_{ij}$ is the additive inverse of x_{ij} in F .

Special Cases

(i) $F = \mathbb{R}$

$$F^{m \times n} = \mathbb{R}^{m \times n}$$

(real-valued $m \times n$ matrices)

(ii) $F = \mathbb{C}$

$$F^{m \times n} = \mathbb{C}^{m \times n}$$

(complex-valued $m \times n$ matrices)

$F^{m \times n}$ is a Vector Space over F

It is also called the vector space of $m \times n$ matrices.

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$S = \{ \text{all real-valued functions defined over } \mathbb{R} \}$.

with Addition and multiplication defined as:-

$$+ : (f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$\cdot : (\alpha f)(x) = \alpha \cdot f(x)$$

(example:-

$$f(x) = x^2 \\ x \in \mathbb{R}$$

$$f(x) = \sin x \\ x \in \mathbb{R}$$

$f_1, f_2 \in S$

Real

Addition

Real multiplication

Claim: S is a vector space over $\mathbb{R}$.

① What is $0 \in S$?

$$0(x) + f(x) = f(x), \quad \forall x \in \mathbb{R}$$

$$0(x) = 0, \quad \forall x$$

② What is $f_{\perp}(x)$?

$$f(x) + f_{\perp}(x) = 0(x), \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f(x) + f(-x) = 0, \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f(-x) = -f(x)$$

$$\textcircled{4} \quad C^{(0)}(\mathbb{R}) = \left\{ f: \mathbb{R} \rightarrow \mathbb{R}, f \text{ is continuous} \right\}$$

$f_1 + f_2$: same as above

αf : " " "

$C^{(0)}(\mathbb{R})$ is a vector space

$$\textcircled{5} \quad \mathbb{F}^\infty = \left\{ (x_1, x_2, x_3, \dots), x_i \in \mathbb{F} \right\}$$

Set of all infinite sequences $(x_i \in \mathbb{F})$

We call them Digital signals if $\mathbb{F} = \mathbb{R}, \mathbb{C}$.

$$+ : (x_1, x_2, \dots) + (y_1, y_2, \dots) = (x_1 + y_1, x_2 + y_2, \dots)$$

Field Addition

- $\alpha \cdot (x_1, x_2, \dots) = (\alpha x_1, \alpha x_2, \dots)$

Field multiplication

Subspace

Given a vector space V (over a field F), a subspace of V is a non-empty subset of V (denoted by W , $W \subseteq V$) with the following properties:-

- (i) If $u, v \in W$, then $u+v \in W$
- (ii) If $u \in W$, then $\alpha \cdot u \in W$, for all $\alpha \in F$

Q Is $0 \in W$? (if W is a subspace)
YES.

Q Is W a vector space?
YES.

By virtue of (i) and (ii) all axioms of a vector space are obeyed by W .

Let's verify some of them:

① Every $v \in W$ has an additive inverse.

$v \in \mathcal{V} \Rightarrow v$ has an additive inverse

Let's denote it by v_I .

$$v_I = 1_I \cdot v$$

$\downarrow$
scalar multiplication

(1_I is additive
inverse of
 $1 \in F$)

If $v \in \mathcal{W}$, then (ii) implies $\alpha \cdot v \in \mathcal{W}$

Choose $\alpha = 1_I$.

$$\Rightarrow 1_I \cdot v \in \mathcal{W}$$

(2) Commutativity of vector addition:

$$\text{If } u, v \in \mathcal{W} \Rightarrow u + v = v + u$$

$$u, v \in \mathcal{W} \Rightarrow u, v \in \mathcal{V} \quad (\mathcal{W} \subseteq \mathcal{V})$$

$$\Rightarrow \boxed{u + v = v + u} \quad (\text{Property of } \mathcal{V} \text{ as a v.s.})$$

Examples

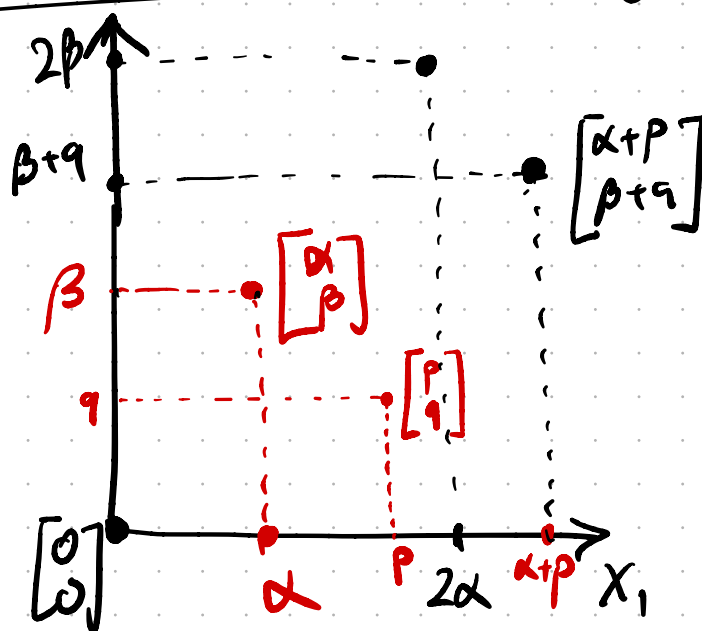
① $\mathcal{V}$ is a v.s. over a field F .

$\mathcal{W} = \{0\}$ is a subspace of $\mathcal{V}$.

SELF-SUFFICIENT SUBSPACE !

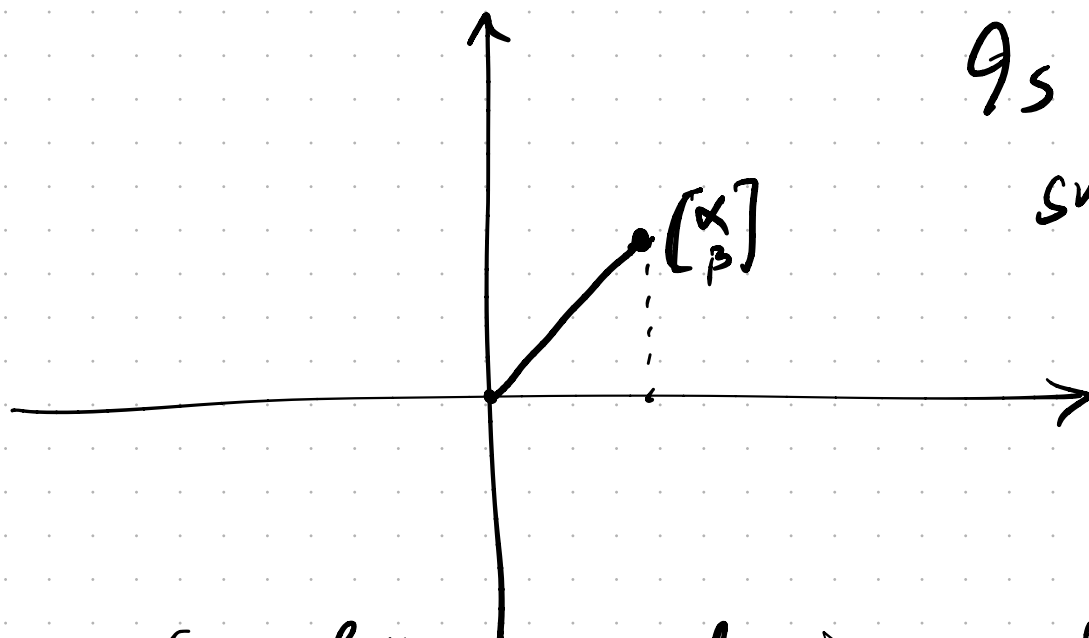
② $F = \mathbb{R}$, $\mathcal{V} = \mathbb{R}^2$

Subspaces of $\mathbb{R}^2$: Geometry



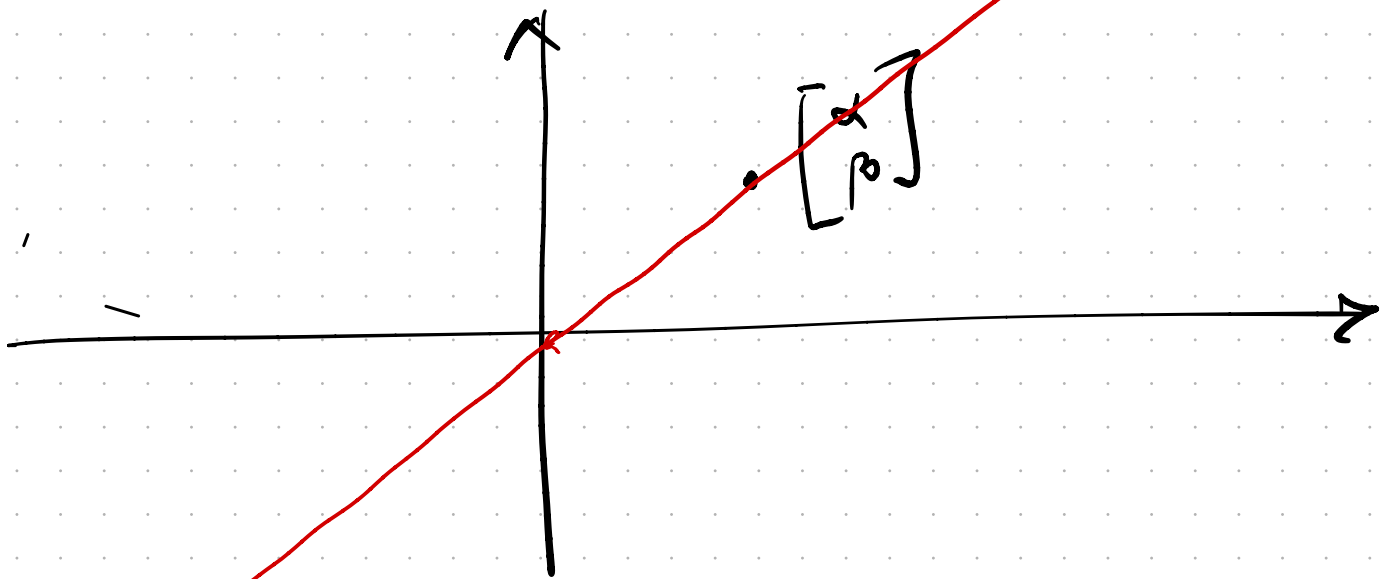
$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} \alpha+p \\ \beta+q \end{bmatrix} \quad (\text{Parallelogram Law})$$

$$\gamma \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \gamma\alpha \\ \gamma\beta \end{bmatrix} \quad (\text{Stretching / Scaling})$$



Is W a
subspace
No!

But the following set is a subspace



Given a vector $v_0 \in \mathbb{R}^2$

$L_{v_0} = \{ \alpha \cdot v_0, \alpha \in \mathbb{R} \}$: line passing
through
origin and
 v_0

Is αv_0 a subspace? It satisfies (ii)

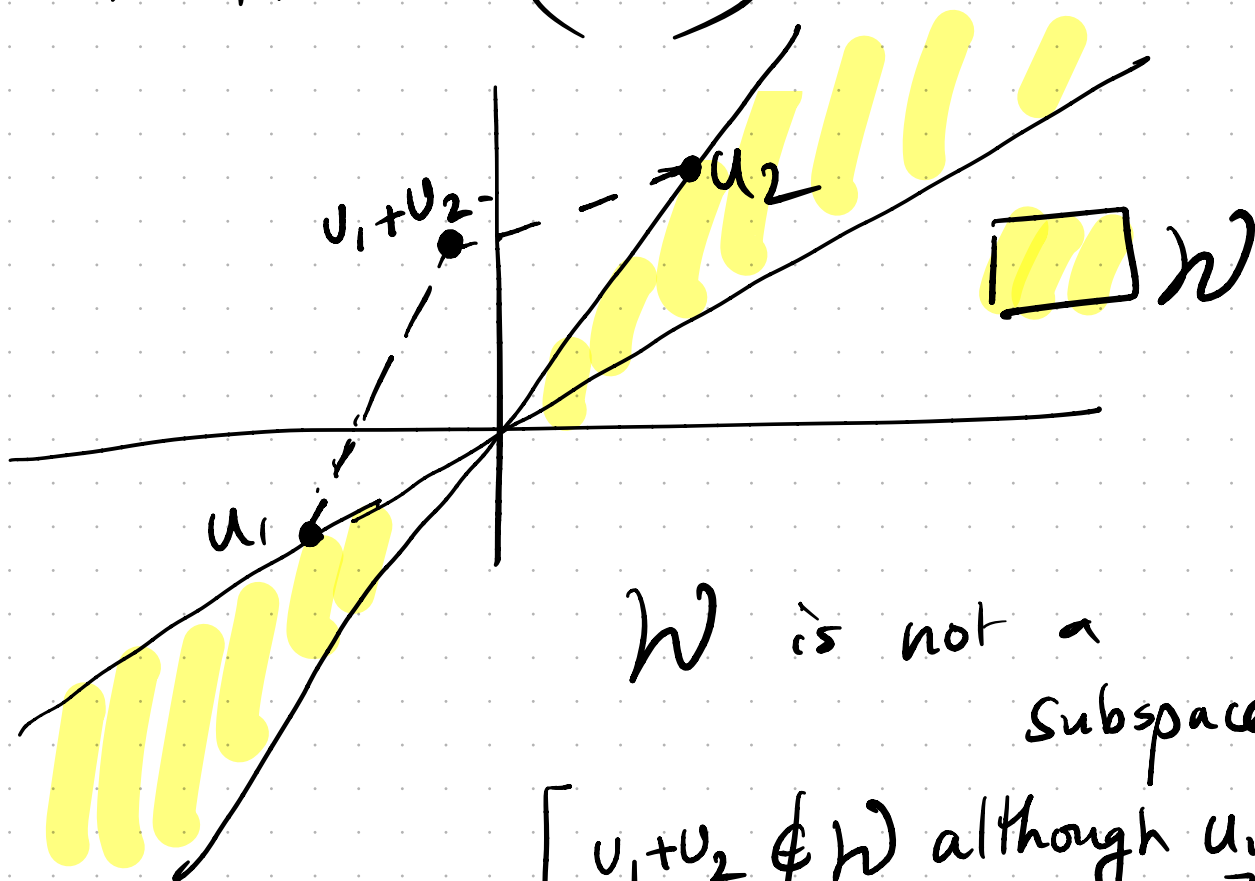
Does it satisfy (i)?

Consider $u_1, u_2 \in \alpha v_0$

$$u_1 = \alpha_1 v_0, \quad u_2 = \alpha_2 v_0$$

$$\Rightarrow u_1 + u_2 = (\alpha_1 + \alpha_2) v_0 = \alpha_3 v_0 \in \alpha v_0$$

(2)



W is not a subspace

$$[u_1 + u_2 \notin W \text{ although } u_1, u_2 \in W]$$

(3) Convince yourself that subspaces of $\mathbb{R}^2$ are of three types -

$$(i) \{0\}$$

(ii) Lines through origin

$$(iii) \mathbb{R}^2$$

NOTE: In $\mathbb{R}^n$, $n \geq 2$, subspaces can be of many additional types. We will formally characterize them soon.

MATRIX VECTOR PRODUCT

Given a matrix $A \in \mathbb{F}^{m \times n}$,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Define the following map from $\mathbb{F}^n$ to $\mathbb{F}^m$

$$f: \mathbb{F}^n \rightarrow \mathbb{F}^m$$

$$f(x) = A \cdot x, \quad \left(x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right)$$

where $y = A \cdot x$ is defined as

$$\begin{aligned} y_1 &= a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n \\ y_2 &= a_{21} \cdot x_1 + a_{22} \cdot x_2 + \dots + a_{2n} \cdot x_n \\ &\vdots \\ y_m &= a_{m1} \cdot x_1 + a_{m2} \cdot x_2 + \dots + a_{mn} \cdot x_n \end{aligned}$$

field Addition (pointing to the '+' sign in the first equation)
field multiplication (pointing to the dot in the first equation)

The above relations are compactly represented as

$$y = A \cdot x$$

And the map is called "MATRIX-VECTOR" PRODUCT.

Two Versatile subspaces of $\mathbb{F}^m$ and

$\mathbb{F}^n$:- Given $A \in \mathbb{F}^{m \times n}$ (A is fixed)

① $\left\{ y \in \mathbb{F}^m, \text{ s.t. there exists } x \in \mathbb{F}^n \text{ satisfying } y = Ax \right\}$

CLAIM: Above is a subspace - and it is called the Range-space of A and denoted by $R(A)$.

Verify using "definition" of above map.

② $\left\{ x \in \mathbb{F}^n, \text{ s.t. } Ax = \underset{\substack{\downarrow \\ 0 \in \mathbb{F}^m}}{0} \right\}$

Claim: This set is a subspace of $\mathbb{F}^n$.
and it is called Null space of A

$(N(A))$ (Verify Yourself)

A Famous Null-space of $\mathbb{R}^n$

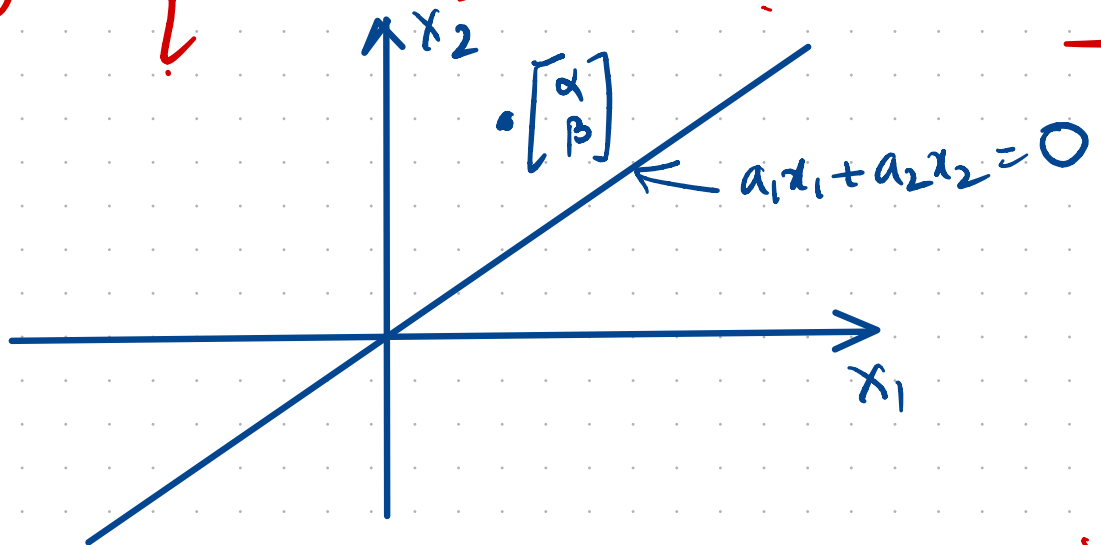
$$A = [a_1 \ a_2 \ \dots \ a_n] \quad (1 \times n \text{ matrix})$$

$$N(A) = \left\{ x \in \mathbb{R}^n, \text{ s.t. } a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0 \right\}$$

This special null space is an example of a hyperplane.

$$n=2$$

$$N(A) = \left\{ x \in \mathbb{R}^2, a_1 x_1 + a_2 x_2 = 0 \right\}$$



With respect to the line $a_1 x_1 + a_2 x_2 = 0$, any point (say $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$) satisfies exactly one of the two relations:
 $a_1 \alpha + a_2 \beta < 0$ or $a_1 \alpha + a_2 \beta \geq 0$

Generalizes to $\mathbb{R}^n$

Consider any point $u \in \mathbb{R}^n$

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

$$N(A) = \left\{ x \in \mathbb{R}^n, \text{ s.t. } a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0 \right\}$$

either $a_1 u_1 + a_2 u_2 + \dots + a_n u_n \leq 0$

OR $a_1 u_1 + a_2 u_2 + \dots + a_n u_n > 0$

